Supplement: Methods and Reproducibility Record

Harun Yahya Gökay

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1 Notation and forecast anchors

I use the notation in Table S1 throughout the paper and supplement. In the main paper's Table 2, an anchor is a risk date for which the target is observed and the compared models all provide an out-of-sample prediction. This common set makes each paired AUC comparison use the same dates.

Table S1: Notation used in the analysis.

Symbol	Meaning
$r_{i,t}$	Continuously compounded return for industry i on date t.
$h_{i,t}$	Filtered log variance for industry i.
$\mathbf{z}_t$	Past-only standardized vector of filtered states.
g_t	Online fitted map from $\mathbf{z}_t$ to $\mathbf{z}_{t+1}$.
$J_t = \mathrm{D}g_t$	Jacobian, or local derivative, of the fitted map.
F_t	Largest singular-value amplification of J_t over the primary horizon.
R_t	Mean rotation of the leading right-singular subspace.
H_t	Effective mode count from the normalized squared singular values.
$\mathbf{S}_t$	Three returns-only stress summaries.
M_0, M_1	Baseline feature sets; M_1 adds AR(1) and absorption ratio to M_0.
$M_0 + F$, $M_1 + F$	Corresponding feature sets with fragility added.
ΔAUC	AUC with F minus AUC for the matching baseline.
Anchor	A forecast risk date with an observed target and predictions for every model in the comparison.

2 Kalman filter and parameter estimation

I use a diagonal Gaussian quasi-likelihood filter for the log-variance state:

$$h_t^- = \mu + \Phi(h_{t-1} - \mu), \qquad P_t^- = \Phi P_{t-1}\Phi^\top + Q, \tag{1}$$

$$\widehat{y}_t = h_t^- + m, \qquad S_t = P_t^- + R_\xi, \tag{2}$$

$$K_t = P_t^- S_t^{-1}, \qquad h_t = h_t^- + K_t(y_t - \widehat{y}_t), \tag{3}$$

$$P_t = (I - K_t)P_t^-(I - K_t)^\top + K_t R_\xi K_t^\top. \tag{4}$$

I symmetrize the covariance after the Joseph update and use a fixed numerical jitter only if a Cholesky factorization requires it. I pass exponentials through a shared clamp; non-finite state or covariance values and state-bound violations stop the run.

The original C++ prototype expressed this update with an unscented transform. Because both transition and observation are linear, its transformed moments equal the Kalman recursions. I use the direct Kalman update in the reported implementation and verified the equivalence numerically to $1.24e-14$.

I estimate μ_i, ϕ_i, and q_i on the estimation window only. I minimize the Gaussian innovation quasi-likelihood for $y_{i,t} = \log(r_{i,t}^2 + c)$, with the log-chi-square mean correction described in the main paper [2]. I do not apply a moment-matching clip or post-fit floor. The zero-as-missing exploratory variant skips both the per-asset likelihood contribution and Kalman measurement update at exact zero returns; the filter still predicts the state and covariance forward on those dates.

3 Online response map and spectral summaries

I set $x_t = [1, \mathbf{z}_t^\top]^\top$ and estimate the response coefficients with recursive least squares and forgetting factor λ:

$$k_t = P_{t-1}x_t/(\lambda + x_t^\top P_{t-1}x_t), \tag{5}$$

$$B_t = B_{t-1} + (\mathbf{z}_{t+1} - B_{t-1}x_t)k_t^\top, \tag{6}$$

$$P_t = (P_{t-1} - k_t x_t^\top P_{t-1})/\lambda. \tag{7}$$

I initialize $P_0 = 10I$, symmetrize each update, and check positive definiteness after burn-in. I use past-only expanding standardization and do not smooth before the primary RLS fit. Centered finite differences recover the linear reference map within $9.77e-11$; they are a validation check, not a derivative of the fitted RLS map.

For singular values s_i of J_t, I set $p_i = s_i^2 / \sum_j s_j^2$ and $H_t = \exp(-\sum_i p_i \log p_i)$. Right-singular vectors describe input perturbations that receive the greatest local amplification; left-singular vectors describe output locations. I use eigenvalues for persistence and singular values for transient amplification in non-normal maps. For non-symmetric J_t, I use Wedin's singular-subspace perturbation bound [4]. For symmetric covariance matrices, I use Davis–Kahan [3].

4 Events, forecasts, and block inference

I compound the equal-weight simple-return portfolio into a wealth index and measure drawdown from its trailing 250-session peak. I merge crossings less than 120 sessions apart. For an onset t_j, the target is one when an onset falls in the next 60 sessions; I remove event-cluster dates from the risk set. I scale each logistic feature using training data only, refit every 252 sessions, and purge the final 60 training observations.

I use 250-session circular blocks in the frozen AUC bootstrap because this is approximately one trading year and exceeds the 120-session event-merging window. This bandwidth preserves longer-run dependence in paired score-label observations; I did not choose it by optimizing the resulting interval. After freeze, I compare 60-, 125-, and 250-session blocks and report all three intervals in Table S5. I use the stationary-observation bootstrap rationale of Künsch [5]. Holm's step-down adjustment controls family-wise error across the two primary AUC comparisons [6].

In the post-freeze refit-aware bootstrap, I draw training blocks available at each original cutoff, refit both M_1 and $M_1 + F$ logistic models, draw blocks from the test period, and recompute AUC levels and their difference. I hold the upstream volatility filter and RLS path fixed, so these intervals capture logistic-fitting and test-block uncertainty but not all state-estimation uncertainty. The fixed-score block-length comparison below holds model scores fixed and isolates interval bandwidth.

5 Frozen signal results and zero-return checks

The pre-test gates appear once in main Table 1; I do not repeat that table here. The frozen pre-onset placebo probabilities are raw $p = (0.385, 0.814, 0.825)$ and Holm-adjusted $p = (1, 1, 1)$ for (R, H, F). The adjusted values are capped at one because each raw probability is large. I do not interpret that correction as evidence that the observed statistics sit at their placebo medians.

I count exact zero log-return cells by period in the ten-industry source. Table S2 reports portfolio-date cells and dates with any zero. For an exact zero, the frozen variance measurement is $\log(0 + c) = \log c = -18.42$ at $c = 10^{-8}$; the adjusted 10^{-6} floor gives -13.82. The calibration period has the highest zero share (0.85%); zeros remain a small fraction of observations in each period. I report two post-freeze treatments: replacing the frozen 10^{-8} variance floor with 10^{-6}, and treating exact zero observations as missing for the per-asset filter update. In both treatments, I keep the original simple returns in the equal-weight drawdown target and stress variables.

6 Factor-aligned direction null

I retain the frozen Haar subspace comparison and add a null that fixes the leading rolling-covariance eigenvector in every draw while sampling the remaining two orthonormal directions in its orthog-

Table S2: Exact zero-return counts in the ten-industry panel.

Period	Dates	Zero cells / cells	Zero share	Dates with a zero
Estimation (1926–1949)	6943	442 / 69430	0.637%	425
Calibration (1950–1969)	5087	433 / 50870	0.851%	419
Test (1970–2026-09)	14308	701 / 143080	0.49%	680

Table S3: Post-freeze zero-return sensitivity. I show Holm-adjusted block-shuffle p values for (R, H, F), the richer-baseline AUC increment, its paired 250-session interval, and the common test sample. F uses 4,999 shuffles; the retained R and H raw probabilities use 499 shuffles and are Holm-adjusted with the new F value.

Treatment	Adjusted p_R, p_H, p_F	ΔAUC	95% interval	Clusters	Anchors
variance floor 1e-6	0.748, 0.98, 0.0804	-0.007123	[-0.0155, 0.00178]	21	10793
zero returns missing	1, 1, 0.0306	-0.0121	[-0.025, 0.000435]	21	10793

onal complement. This asks whether the observed susceptibility space aligns unusually well after accounting for the market's leading covariance factor. I calculate date-wise angle percentiles and use circular 250-session blocks to summarize their dependence. I estimate the effective sample size from the integrated autocorrelation time rather than treating daily dates as independent. For the frozen Haar comparison, the median date-wise percentile is median 5.9% (95% block interval [3.5%, 8.2%]); effective sample size 49.69 of 14308 dates. For the factor-aligned null, the median percentile and interval are 18.1% (95% block interval [13.9%, 25%]); effective sample size 48.06 of 14308 dates.

7 H2 levels, alerts, and sensitivity

For the frozen target and default feature definitions, I obtain a no-skill AUC of 0.5, an EWMA-volatility-only AUC of 0.748 with interval [0.659, 0.829], an M_1 AUC of 0.741 with interval [0.646, 0.842], and an $M_1 + F$ AUC of 0.732 with interval [0.639, 0.832]. The paired increment is -0.00882 with interval [-0.0227, 0.00292] across 10793 common anchors.

At a fixed 5% alert rate, M_1 precision and recall are 0.606 and 0.277; $M_1 + F$ precision and recall are 0.587 and 0.268; the EWMA-only values are 0.396 and 0.181. I use the same alert-rate rule for all three models.

I fit the full post-freeze grid over drawdown threshold, forecast horizon, and logistic penalty C. Table S4 reports the event count, common anchors, AUC levels, incremental AUC, paired 250-session interval, and EWMA AUC for each setting. These estimates are exploratory and I report non-estimable configurations explicitly.

Table S4: Post-freeze H2 sensitivity grid. Each row changes the drawdown threshold, forecast horizon, and C; intervals are paired circular-block intervals.

Drawdown	Horizon	C	Events	Anchors	AUC M_1	AUC $M_1 + F$	ΔAUC	95% interval	EWMA AUC
-10%	20	0.1	23	9466	0.611	0.603	-0.00848	[-0.0369, 0.0224]	0.589
-10%	20	1	23	9466	0.602	0.596	-0.00599	[-0.0397, 0.0273]	0.591
-10%	20	10	23	9466	0.601	0.596	-0.00564	[-0.0337, 0.0237]	0.591
-10%	60	0.1	23	9426	0.476	0.459	-0.0174	[-0.0566, 0.0142]	0.469
-10%	60	1	23	9426	0.476	0.459	-0.0171	[-0.0521, 0.0168]	0.469
-10%	60	10	23	9426	0.476	0.459	-0.0171	[-0.0557, 0.0159]	0.47
-10%	120	0.1	23	9366	0.485	0.461	-0.0238	[-0.0601, 0.0108]	0.468
-10%	120	1	23	9366	0.485	0.462	-0.0237	[-0.0649, 0.0122]	0.469

Drawdown	Horizon	C	Events	Anchors	AUC M_1	AUC $M_1 + F$	ΔAUC	95% interval	EWMA AUC
-10%	120	10	23	9366	0.485	0.462	-0.0237	[-0.0611, 0.00744]	0.469
-15%	20	0.1	21	10833	0.862	0.865	0.00273	[-0.00994, 0.0164]	0.839
-15%	20	1	21	10833	0.851	0.853	0.00276	[-0.0102, 0.016]	0.839
-15%	20	10	21	10833	0.849	0.852	0.00262	[-0.0109, 0.0174]	0.839
-15%	60	0.1	21	10793	0.743	0.734	-0.00867	[-0.0211, 0.00376]	0.747
-15%	60	1	21	10793	0.741	0.732	-0.00882	[-0.0217, 0.00308]	0.748
-15%	60	10	21	10793	0.741	0.732	-0.00886	[-0.0217, 0.00512]	0.748
-15%	120	0.1	21	10733	0.67	0.66	-0.0102	[-0.0263, 0.00627]	0.725
-15%	120	1	21	10733	0.669	0.659	-0.0101	[-0.0258, 0.00577]	0.725
-15%	120	10	21	10733	0.669	0.659	-0.0101	[-0.0261, 0.00764]	0.725
-20%	20	0.1	10	12767	0.882	0.871	-0.0112	[-0.0615, 0.0264]	0.777
-20%	20	1	10	12767	0.876	0.853	-0.0231	[-0.0706, 0.015]	0.776
-20%	20	10	10	12767	0.876	0.835	-0.0405	[-0.122, 0.0142]	0.776
-20%	60	0.1	10	12727	0.818	0.817	-0.00108	[-0.00976, 0.00641]	0.678
-20%	60	1	10	12727	0.819	0.818	-0.00123	[-0.00949, 0.00643]	0.679
-20%	60	10	10	12727	0.82	0.818	-0.00126	[-0.00918, 0.00654]	0.679
-20%	120	0.1	10	12667	0.709	0.723	0.0135	[-0.0202, 0.0467]	0.665
-20%	120	1	10	12667	0.711	0.724	0.0131	[-0.0189, 0.0467]	0.666
-20%	120	10	10	12667	0.711	0.724	0.0131	[-0.0178, 0.0497]	0.666

Table S5: H2 interval sensitivity to the circular-block length. I use the same common anchors and fixed expanding-window scores for all rows.

Block length	AUC M_1	AUC $M_1 + F$	ΔAUC	95% interval
60	0.741	0.732	-0.00882	[-0.0189, 0.00228]
125	0.741	0.732	-0.00882	[-0.0202, 0.00346]
250	0.741	0.732	-0.00882	[-0.0214, 0.00501]

The separate refit-aware analysis used 399 successful 250-session resamples. Its 95% intervals are [0.592, 0.837] for AUC M_1, [0.586, 0.829] for AUC $M_1 + F$, and [-0.0427, 0.0168] for their difference. I hold the filter and response-map paths fixed as described above.

The intervals reported around the rounded point estimate $\Delta\text{AUC} = -0.00882$ differ by design. The frozen interval is Holm-adjusted and uses 1,000 250-session bootstrap replicates (seed 20261108); Table S4 reports unadjusted fixed-score intervals from 399 setting-specific replicates; Table S5 uses 399 fixed-score replicates with a seed that varies by block length. The refit-aware interval uses 399 replicates (seed 20261119) and refits the logistic models, so it captures a different source of uncertainty.

8 Subperiod and forgetting-factor checks

I compare the observed median F with its block-shuffle null separately for 1970–1999 and 2000–2026. I adjust the two raw probabilities with Holm's procedure. Table S6 reports both periods even if either test is inconclusive.

Table S6: Post-freeze subperiod fragility surrogate results. Each F test uses 4,999 block shuffles; Holm adjustment is across the two periods.

Period	Observed median F	Null median	Raw p	Holm p
1970–1999	1.06	1.09	0.087	0.087
2000–2026	1.10	1.14	0.0366	0.0732

I also recompute the full R, H, and F summaries and their block-shuffle probabilities for $\lambda \in \{0.990, 0.995, 0.998, 0.999\}$ while keeping parameters fixed at the primary estimation values. Table S7 reports medians and Holm-adjusted probabilities for each value. Each F test uses 4,999 block shuffles; the retained R and H raw probabilities use 499 shuffles and their Holm adjustments are recomputed with the new F probability.

Table S7: Forgetting-factor sensitivity with primary estimation parameters held fixed. Adjusted probabilities are ordered (R, H, F); F uses 4,999 draws and R, H use retained 499-draw raw probabilities.

λ	Median R [IQR]	Median H [IQR]	Median F [IQR]	Holm-adjusted p_R, p_H, p_F
0.990	0.363 [0.278, 0.476]	9.78 [9.74, 9.82]	1.36 [1.23, 1.5]	0.27, 0.204, 0.0024
0.995	0.232 [0.173, 0.325]	9.92 [9.9, 9.93]	1.2 [1.13, 1.28]	0.92, 0.124, 0.0078
0.998	0.13 [0.0934, 0.188]	9.97 [9.97, 9.98]	1.08 [1.04, 1.13]	0.872, 0.962, 0.003
0.999	0.0718 [0.0517, 0.103]	9.99 [9.98, 9.99]	1.03 [1.01, 1.07]	0.416, 0.502, 0.0006

Table S9: Frozen robustness summaries. I report the EMA-before-RLS and $\tau = 20$ rows along with the primary and alternate forgetting-factor summaries.

Specification	Median R	Median H	Median F
Primary, lambda 0.998	0.1303	9.974	1.08
Alternate lambda 0.995	0.2319	9.919	1.199
EMA before RLS	0.198	9.99	1.254
Horizon tau 20	–	–	1.251

Table S10: Definitions and fixed thresholds for the validation gates.

Gate	Purpose	Pass rule
U	Numerical equivalence and local-Jacobian checks	UKF/Kalman error $\leq 1 \times 10^{-10}$; finite-difference error $\leq 1 \times 10^{-9}$
E1	Variance-parameter recovery across assets	At least 8/10 assets; $\lvert\Delta\phi\rvert \leq 0.05$; q ratio ≤ 2; no box-boundary estimate
G1	Recovery of a planted heterogeneous transition map	Relative Frobenius error ≤ 0.35; off-diagonal correlation ≥ 0.5
G2	Separation of rotating and amplifying synthetic regimes	Rotation contrast ≥ 0.15 radians; log-amplification contrast ≥ 0.15; stationary rotation $\leq 1 \times 10^{-10}$
G3	Held-out filtering improvement over a constant-state comparator	Relative RMSE improvement $\geq 5\%$
G4	Historical boundedness and direct-RLS comparison	Median $\rho \leq 1.05$; median $F_5 \leq 3$; maximum $F_5 \leq 1000$; direct-RLS relative gap ≤ 0.5

10 Seventeen-industry aggregation check

I downloaded the daily 17-industry series from the same U.S. stock universe and used it as an aggregation check. I find 22 test-period event clusters in the 17-industry aggregation and match 19 to the 21 ten-industry events within 120 sessions (86.4% of the 17-industry events). Its F surrogate comparison gives raw $p = 0.002$ from 499 draws. I estimate ΔAUC for $M_1 + F$ versus M_1 as -0.0237 (95% interval [-0.052, -0.00121]) with 10288 anchors. I do not count this as an independent replication or as a meaningful increase in H2 event power. I found no daily developed ex-U.S. industry portfolio series in the French Library's industry tables; its available daily developed ex-U.S. portfolios are sorted by size and book-to-market, not by industry.

11 Deviations and amendments

I record the following implementation corrections and access deviations in my development log and protocol-amendment record. None changed a frozen threshold, statistic, or verdict.

G1 fixture correction. My initial synthetic map used equal off-diagonal coefficients, so the declared off-diagonal correlation had no meaningful ground-truth variation. I replaced it with a heterogeneous fixture; after the first design remained too collinear, I used small signed, zero-centered off-diagonal coefficients around a 0.95 diagonal. I kept the thresholds and candidate forgetting factors fixed.

Event-placebo indexing. My first complete analysis invocation stopped before writing results because the event-aligned profile code referenced an onset outside its loop scope. I corrected the offsets to index from each cluster's first onset. The placebo statistic, test-period dates, and inference rules did not change.

Alternate robustness row. My initial robustness summary reused the selected forgetting factor as the alternate because the selected value was the largest candidate. I corrected the row to the other prespecified value, $\lambda = 0.995$, and recomputed its summary. The primary H1 and H2 results are unchanged.

CMake compiler identification. The Visual Studio generator omitted compiler ID and version from its cache. I recorded MSVC 19.50.35721.0 and MSBuild 18.0.5, with executable hashes, in the build-provenance correction record.

Final-row preview. Before the freeze, my initial filesystem inventory displayed the final rows of the source archive. I disclose this access deviation; no test-period aggregate or model result was computed from that preview.

Early lambda harness. An early gate-harness version loaded the pre-test panel before completing synthetic lambda selection. I did not analyze a test-period aggregate, label, AUC, or model result. The final runner completes synthetic selection first and verifies the frozen protocol hash before reading derived returns.

12 Reproducibility record

I preserve the frozen protocol, analysis plan, validation, and study result. The exact SHA-256 for `results/study.json` is included in Table S11; that file contains derived states, diagnostics, model estimates, and summaries, not raw French Library return rows. I record the source archive, extracted member, derived log-return panel, frozen study, the exploratory plan and three hashed extensions, the protocol, precision-run outputs, amendment log, frozen executable, exploratory C++ source and binary, and exploratory analysis script hashes below. I do not include raw source archives in the repository.

I built the implementation with MSVC 19.50.35721.0and C++ C++17, using CMake cmake version 4.2.1and pybind11 3.0.4. The Python environment used Python 3.12.0, NumPy 2.3.5, and SciPy 1.16.3. I verified a byte-for-byte rebuild of the core. The F precision extension uses its own hashed plan and runner, and its outputs are listed above. I include clean-clone commands in the repository README.

I use a 250-session block for the primary bootstrap because it spans about a trading year and exceeds the 120-session event-merging window; I report shorter block-length checks rather than selecting a more favorable bandwidth. The frozen verdicts remain unchanged in every post-freeze report.

Table S11: Source and analysis artifact SHA-256 values.

Artifact	SHA-256
Source archive	4c79bad80b6f4eee9d08e649ce223d1c51aadd65043b65baca50b54981d38d2501cfe3d5e54f6ddfcc3dc84fb793f10f
Extracted member	6b0218fe59f780969f4dca84e72e823e84f596454c562a0e299068baf8ebdfde
Derived returns	e35f2984aed979c6b77f3bd9900338d5732f6b8b15661ec9b176489c989b5ccb
Frozen study JSON	a4c09d453d24597aee78c430cdc0d0ae7ea3f2c36f7150e3a13478dbfe6b7205
Analysis plan	e31d0934132d04eb6087935fe5688388f6b3ebc6ed9598f5f3b88c8e742ba4ea
Post-freeze plan	7e2682245d04d757dc045ec778872647 6fd373bd60d5857abe54ce6066fa2a07
Block-length plan	3c37906422c85c2d09b4bd400be55338d4bf5aeb661a07860210f01fc384be0a
Haar dependence plan	25d8c4119d186ba018bb579c16f65ee6311a60d6715f6fe5551df3b8e4cd2f0f
F precision plan	213d3717e46fbc877a53ac64448731add41ea3e016e8134629a0508b18212fb5
F precision runner	4bf321ea2e0d3bae833b9eb956123a2300e87447b032b375f2720deebbf89a09
F primary precision output	21d8436445122608e9bd3f00c14788cb680477b57b1781a21916d3ac704f7bb9
Zero-variant precision output	1923ddca1f641c7194264b88bd18805f9a97dd778315972ddcd1713e89ea1dc2
Subperiod precision output	fba8f87e633cc7df99696c1ed7a0a29734541764647afdc196e126f3cd94458c
Lambda-grid precision output	fc354ff20874e02b9ccffc9ddf8d4e02353db57bd795a5c2bb31e2830ea56f61
F precision comparison	1451247116aa0c171fc3facaa4f6320e246b0e9328a66b91a0238d400a8e01ff
Protocol amendment log	093b836e0515a3a8acc1f9771ec991c5a860e120e56490bcfe14675241d33f59
Protocol	58929c0e78747a8da334c79c7fa311c0c20b1ccb5bc22b7137938854499f81c7
17-industry source archive	4a88694e95af224bf8d0189c0f846f343e30b7e2a5ef30c42c090fba656d023d
17-industry extracted member	078fe96e092c6517332b58bd143e0cd7cd1b005e90cb6beddd40e3b50c23c180
17-industry derived returns	41a4d1ed1216be6fe79e9e416057ee9bb0057ea8aa410408ff2ffad3d72940b0
Frozen core executable	8687d08bcf89b47f899f9acb71fca3de873d59d924e9711be31a6d2ddfd56dcd
Exploratory core source	84402059b4cd4cc4d71badf33e303eb873bfa47c9347ed3d0148492d9bdf1619
Exploratory core executable	4da33d1df68ec51a827c7712b44b741b5d03c62d1f7ab342cc6697fa426431f0
Exploratory analysis script	c965ea7f1ca62f66256d9d52d2eca9c2

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